

DSA Benchmark Studio

Benchmark Export Report

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Total Benchmarks: 9

Algorithm	Category	Dataset	Size	Average (ms)	Peak Memory	Time Complexity
merge_sort	Sorting	random	1000	4.1808	0.0	$O(n \log n)$
insertion_sort	Sorting	random	1000	47.5074	0.0	$O(n^2)$
selection_sort	Sorting	random	1000	51.9222	0.0	$O(n^2)$
counting_sort	Sorting	random	1000	0.9195	0.0	$O(n+k)$
quick_sort	Sorting	random	1000	3.8801	0.0	$O(n \log n)$
shell_sort	Sorting	random	1000	2.0682	0.0	Depends on gap sequence
insertion_sort	Sorting	random	1000	22.6737	0.0	$O(n^2)$
heap_sort	Sorting	random	1000	0.3551	0.0	$O(n \log n)$
quick_sort	Sorting	random	1000	1.6663	0.0	$O(n \log n)$

Detailed Benchmark Results

merge_sort

Category: Sorting
Dataset Type: random
Dataset Size: 1000
Runs: 3
Average Time: 4.180766666650015 ms
Median Time: 4.014699999970617 ms
Minimum Time: 3.30740000026708 ms
Maximum Time: 5.220199999712349 ms
Std Deviation: 0.9671528125409968 ms
Peak Memory: 0.0 KB
Time Complexity: $O(n \log n)$
Space Complexity: $O(n)$
Stable: True
In Place: False

insertion_sort

Category: Sorting
Dataset Type: random
Dataset Size: 1000
Runs: 3
Average Time: 47.50743333321831 ms

Median Time: 49.10959999961051 ms
Minimum Time: 40.91059999973368 ms
Maximum Time: 52.50210000031075 ms
Std Deviation: 5.959523603948412 ms
Peak Memory: 0.0 KB
Time Complexity: $O(n^2)$
Space Complexity: $O(1)$
Stable: True
In Place: True

selection_sort

Category: Sorting
Dataset Type: random
Dataset Size: 1000
Runs: 3
Average Time: 51.922166666372505 ms
Median Time: 51.985899999635876 ms
Minimum Time: 41.73660000014934 ms
Maximum Time: 62.0439999993323 ms
Std Deviation: 10.153850015389763 ms
Peak Memory: 0.0 KB
Time Complexity: $O(n^2)$
Space Complexity: $O(1)$
Stable: False
In Place: True

counting_sort

Category: Sorting
Dataset Type: random
Dataset Size: 1000
Runs: 3
Average Time: 0.9194999996301098 ms
Median Time: 1.0360999995100428 ms
Minimum Time: 0.58069999977306 ms
Maximum Time: 1.1416999996072263 ms
Std Deviation: 0.29812232377412456 ms
Peak Memory: 0.0 KB
Time Complexity: $O(n+k)$
Space Complexity: $O(n+k)$
Stable: True
In Place: False

quick_sort

Category: Sorting
Dataset Type: random
Dataset Size: 1000
Runs: 3
Average Time: 3.8800666664731884 ms
Median Time: 2.3172000001068227 ms
Minimum Time: 1.9664000001284876 ms
Maximum Time: 7.356599999184255 ms
Std Deviation: 3.0158710466604064 ms
Peak Memory: 0.0 KB
Time Complexity: $O(n \log n)$
Space Complexity: $O(\log n)$
Stable: False

In Place: True

shell_sort

Category: Sorting
Dataset Type: random
Dataset Size: 1000
Runs: 3
Average Time: 2.0682000000912617 ms
Median Time: 2.139299999726063 ms
Minimum Time: 1.4663000001746695 ms
Maximum Time: 2.599000000373053 ms
Std Deviation: 0.5696873968005663 ms
Peak Memory: 0.0 KB
Time Complexity: Depends on gap sequence
Space Complexity: $O(1)$
Stable: False
In Place: True

insertion_sort

Category: Sorting
Dataset Type: random
Dataset Size: 1000
Runs: 3
Average Time: 22.67366666668143 ms
Median Time: 23.76689999982773 ms
Minimum Time: 18.222599999717204 ms
Maximum Time: 26.031500000499364 ms
Std Deviation: 4.017598679071931 ms
Peak Memory: 0.0 KB
Time Complexity: $O(n^2)$
Space Complexity: $O(1)$
Stable: True
In Place: True

heap_sort

Category: Sorting
Dataset Type: random
Dataset Size: 1000
Runs: 3
Average Time: 0.35509999967568245 ms
Median Time: 0.3569999998944695 ms
Minimum Time: 0.3448999996180646 ms
Maximum Time: 0.3633999995145132 ms
Std Deviation: 0.009395211528523054 ms
Peak Memory: 0.0 KB
Time Complexity: $O(n \log n)$
Space Complexity: $O(1)$
Stable: False
In Place: True

quick_sort

Category: Sorting
Dataset Type: random
Dataset Size: 1000
Runs: 3

Average Time: 1.66633333256868 ms
Median Time: 1.5897000002951245 ms
Minimum Time: 1.4604999996663537 ms
Maximum Time: 1.9487999998091254 ms
Std Deviation: 0.2530093344631316 ms
Peak Memory: 0.0 KB
Time Complexity: $O(n \log n)$
Space Complexity: $O(\log n)$
Stable: False
In Place: True